

Weather-Based Solar Energy Prediction Using Random Forest Regression and Planning System Using Generative AI

1st Al Ameen H

*Dept of Information Technology,
Dhanalakshmi Srinivasan College
of Engineering and Technology
Mamallapuram, Chennai, Tamil Nadu
alameenofficial11@gmail.com*

2nd Hemachandiran J

*Dept. of Information Technology
Dhanalakshmi Srinivasan College
of Engineering and Technology
Mamallapuram, Chennai, Tamil Nadu
hemachandiranj.it2022@dscet.ac.in*

3rd Ranjitha

*Asst. Professor, Dept. of IT
Dhanalakshmi Srinivasan College
of Engineering and Technology
Mamallapuram, Chennai, Tamil Nadu
ranjitha.it@dscet.ac.in*

4th Ravi Kumar

*HOD, Dept. of IT
Dhanalakshmi Srinivasan College
of Engineering and Technology
Mamallapuram, Chennai, Tamil Nadu
ravikumar.ithod@dscet.ac.in*

Abstract—The rapid expansion of rooftop solar photovoltaic (PV) installations in India demands robust energy yield prediction tools that empower residential consumers to make data-driven investment decisions. This paper introduces SolarPredict, a full-stack web-based decision support system that synergises machine-learning-driven solar energy forecasting with real-time weather integration, comprehensive financial analysis, and Generative AI-powered advisory services. The prediction engine employs a Random Forest Regressor trained on 136,476 operational records collected from two solar power plants, achieving a coefficient of determination (R^2) of 0.853 and a normalised mean absolute error (MAE) of 0.032 on a temporally partitioned test set. The planning subsystem leverages Google Gemini, a Generative AI large language model, to deliver personalised natural language recommendations encompassing solar potential assessments, financial guidance, and smart appliance scheduling. The system architecture follows a three-tier microservice paradigm comprising a React frontend, an Express.js API gateway, and a FastAPI-based machine learning inference service, with MongoDB for persistent storage. The platform integrates the OpenWeather API for real-time meteorological data acquisition and incorporates Indian government subsidy calculations under the PM Surya Ghar scheme. Experimental evaluation demonstrates that the proposed system delivers reliable energy predictions with actionable financial and planning insights, thereby enabling homeowners to make informed solar adoption decisions.

Index Terms—solar energy prediction, random forest regression, generative AI, decision support system, machine learning, photovoltaic, weather-based forecasting, renewable energy, smart planning

I. INTRODUCTION

India has committed to an ambitious target of achieving 500 GW of renewable energy capacity by 2030 under its Nationally Determined Contributions (NDC) to the Paris Agreement [1]. Rooftop solar photovoltaic (PV) installations

constitute a significant opportunity for decentralised energy generation, particularly within the residential sector. The Indian government's PM Surya Ghar: Muft Bijli Yojana scheme, inaugurated in 2024, provides subsidies of up to INR 78,000 for residential rooftop solar installations with the objective of solarising one crore households [2]. Despite these incentives, prospective adopters face considerable challenges in accurately estimating energy yields, evaluating financial returns, and comprehending the influence of local weather conditions on solar panel performance.

Accurate solar energy prediction is inherently complex owing to its dependence on multiple meteorological variables, including solar irradiance, ambient temperature, cloud cover, humidity, and wind speed [3]. While physics-based clear-sky models provide theoretical irradiance estimates, they frequently fail to capture the stochastic nature of real-world atmospheric conditions. Machine learning approaches, particularly ensemble methods such as Random Forest, have demonstrated superior performance in modelling these nonlinear relationships between weather parameters and energy output [4].

Furthermore, the emergence of Generative AI technologies, particularly large language models (LLMs), presents new opportunities for creating intelligent planning systems that can translate complex technical predictions into actionable, human-readable recommendations [11]. By integrating Generative AI with ML-based prediction, systems can provide not only accurate forecasts but also personalised guidance on installation planning, financial optimisation, and energy management strategies.

Existing solutions in the literature predominantly focus on either prediction accuracy or system integration, but rarely address the complete decision-making pipeline from weather.

data acquisition through energy prediction to financial analysis and AI-driven planning recommendations. Furthermore, most systems are designed for utility-scale installations and lack the user-friendly interfaces required for residential consumers [5].

This paper presents SolarPredict, an end-to-end web-based prediction and planning system that addresses these gaps through the following contributions:

- A Random Forest Regressor trained on real-world solar plant data [12] with nine engineered features, including cyclical time encodings and synthesised weather parameters for accurate energy prediction.
- Real-time weather data integration via the OpenWeather API for location-specific prediction generation.
- Comprehensive financial analysis encompassing installation costs, government subsidies under the PM Surya Ghar scheme, return on investment (ROI), and 25-year net savings projections.
- A Generative AI-powered planning system utilising Google Gemini LLM for personalised solar advisory, automated summary generation, interactive question answering, and smart appliance scheduling recommendations.
- A complete microservice-based implementation with JWT-authenticated user accounts and prediction history tracking for longitudinal performance analysis.

The remainder of this paper is organised as follows: Section II reviews related work in solar prediction and AI-based planning systems. Section III describes the system architecture and design. Section IV presents the machine learning model development. Section V elaborates on the Generative AI planning system and decision support features. Section VI provides experimental results and analysis. Section VII concludes with a discussion of contributions and future research directions.

II. RELATED WORK

A. Solar Energy Prediction Methods

Solar energy prediction methods can be broadly classified into physical models, statistical models, and machine learning models [3], [14], [16]. Physical models, such as the Hottel clear-sky model, compute solar irradiance based on atmospheric transmittance, solar geometry, and geographic parameters. While theoretically grounded, these models exhibit limitations in handling cloud-induced variability and localised atmospheric effects [6].

Statistical approaches, including Autoregressive Integrated Moving Average (ARIMA) and exponential smoothing, have been applied to solar time series forecasting [15]. However, their inherent assumption of linear relationships constrains their ability to capture the complex nonlinear interactions between weather parameters and energy output [7], [18].

Machine learning methods have gained prominence owing to their capacity to learn nonlinear mappings directly from data. Support Vector Regression (SVR), Artificial Neural Networks (ANN), and ensemble methods such as Random Forest and Gradient Boosting have been extensively studied

[4], [19]. Deep learning approaches, particularly Long Short-Term Memory (LSTM) recurrent neural networks, have also shown promise for temporal solar power forecasting [13], [17]. Among these, Random Forest has demonstrated consistent and robust performance across diverse datasets due to its inherent resistance to overfitting, ability to handle heterogeneous feature types, and interpretable feature importance measures [8].

B. Decision Support Systems for Solar Energy

Several web-based tools exist for solar energy estimation. Google's Project Sunroof employs aerial imagery and 3D modelling to estimate rooftop solar potential [9]. The National Renewable Energy Laboratory's (NREL) PVWatts Calculator provides physics-based estimation of PV system output [10]. However, these tools are primarily designed for the US market and lack the financial models specific to Indian subsidy schemes and electricity tariff structures. Recent studies have highlighted the significant rooftop solar PV potential across Indian states, further underscoring the need for India-specific decision support tools [20].

C. Generative AI in Energy Systems

Recent advances in Generative AI have opened new frontiers in energy system management. Large language models have been explored for natural language interaction with energy data, automated report generation, and explainable AI in energy forecasting [11]. However, the integration of Generative AI as a planning and advisory layer within solar energy prediction systems remains largely unexplored. Our contribution extends this emerging research direction by combining Random Forest-based prediction with a Gemini LLM-powered planning system in a unified, India-focused decision support platform that provides both quantitative predictions and qualitative advisory services.

III. SYSTEM ARCHITECTURE AND DESIGN

A. Overall Architecture

SolarPredict follows a three-tier microservice architecture as illustrated in Fig. 1. The system comprises three principal components:

- 1) **Frontend Layer:** A React 18 single-page application built with Vite, featuring 13 modular components for weather display, prediction visualisation, financial analysis, AI advisory, and prediction history management. Recharts is utilised for interactive data visualization including bar charts, area plots, and comparative dashboards.
- 2) **API Gateway Layer:** An Express.js server that orchestrates communication between the frontend, external APIs (OpenWeather, Google Gemini), the ML inference service, and the MongoDB database. It implements JWT-based authentication, rate limiting (100 requests per 15 minutes), CORS configuration, and request validation middleware.
- 3) **ML Inference Service:** A Python FastAPI application that loads the trained Random Forest model via joblib

deserialization and serves predictions through RESTful endpoints. It accepts panel capacity, geographic coordinates, and weather parameters, returning daily, monthly, and yearly energy estimates with associated confidence metrics.

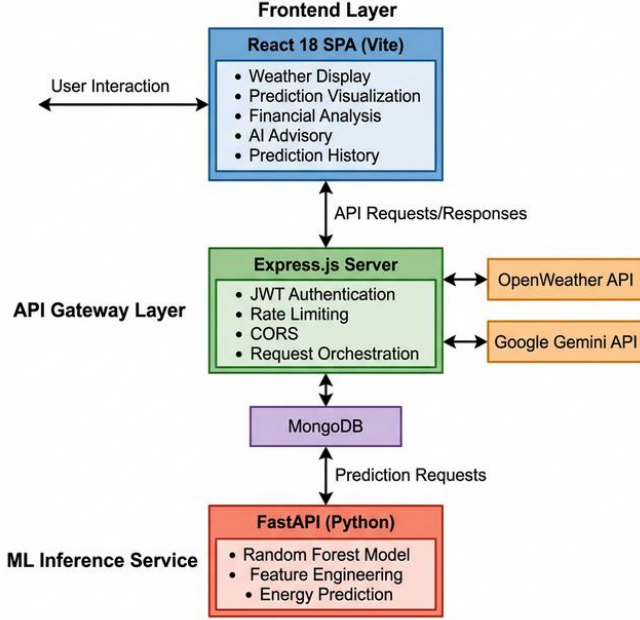


Fig. 1. Three-tier microservice architecture of SolarPredict. The React frontend communicates with the Express.js API gateway, which orchestrates the ML inference service, Generative AI planning service, external weather APIs, and persistent MongoDB storage.

B. Data Flow Pipeline

The end-to-end prediction and planning pipeline operates through the following sequence:

- 1) The user inputs their city name and solar panel capacity (kW) through the React frontend interface.
- 2) The Express.js backend invokes the OpenWeather API to retrieve current weather conditions and a 5-day forecast for the specified location.
- 3) Weather parameters (temperature, humidity, cloud cover percentage, wind speed), along with panel capacity and geographic latitude, are forwarded to the FastAPI ML service.
- 4) The ML service constructs 96 feature vectors (one per 15-minute interval over 24 hours) for each forecast day, incorporating cyclical time encodings and physics-informed solar geometry calculations.
- 5) The Random Forest model predicts normalised power output for each interval, which is subsequently scaled by panel capacity and a system loss factor of 82% to produce energy estimates in kWh.
- 6) The API gateway augments predictions with financial analysis (costs, subsidies, ROI), then forwards the complete context to the Google Gemini Generative AI

service for natural language summary generation and planning recommendations.

- 7) The consolidated response—comprising predictions, financial analysis, AI advisory, and appliance scheduling recommendations—is transmitted to the frontend for interactive visualisation.

C. Technology Stack

Table I summarises the technology stack employed across the system layers.

TABLE I
TECHNOLOGY STACK SUMMARY

Layer	Technologies
Frontend	React 18, Vite, Recharts, Axios
Backend API	Node.js, Express.js, Mongoose, JWT
ML Service	Python 3.x, FastAPI, scikit-learn, joblib
Gen AI Planning	Google Gemini 2.0 Flash LLM
Database	MongoDB (Atlas)
External APIs	OpenWeather API v2.5

IV. MACHINE LEARNING MODEL

A. Dataset Description

The prediction model is trained on real-world operational data obtained from the “Solar Power Generation Data” dataset published on Kaggle [12]. The dataset comprises generation and weather sensor measurements from two solar power plants, organised into four CSV files: Plant_1_Generation_Data.csv, Plant_1_Weather_Sensor_Data.csv, Plant_2_Generation_Data.csv, and Plant_2_Weather_Sensor_Data.csv. The generation data contains timestamped AC and DC power output readings per inverter, while the weather data includes ambient temperature, module temperature, and solar irradiation measurements. The combined dataset encompasses 136,476 records across 44 unique inverters spanning a temporal range from May to June 2020.

B. Feature Engineering

Nine features are systematically engineered from the raw data to capture the principal physical and temporal factors governing solar energy generation:

- 1) **Irradiation:** Direct measurement of solar irradiance (kW/m²) from on-site weather sensors, serving as the primary surrogate for incident solar energy.
- 2) **Ambient Temperature:** Measured atmospheric air temperature (°C) at the plant site, influencing photovoltaic cell efficiency through temperature-dependent semiconductor behaviour.
- 3) **Humidity:** Synthesised proxy variable inversely correlated with temperature and irradiation, representing atmospheric moisture content that attenuates solar radiation through absorption and scattering.

- 4) **Cloud Cover:** Derived as the inverse of measured irradiation relative to the 99th percentile clear-sky irradiance reference, capturing the fractional sky obscuration.
- 5) **Wind Speed:** Generated proxy variable with temperature-correlated convective variation and stochastic Gaussian noise, modelling the cooling effect on panel surface temperature.
- 6) **Hour Sine and Cosine:** Cyclical encoding of the hour of day using $\sin(2\pi h/24)$ and $\cos(2\pi h/24)$, eliminating The artificial discontinuity at midnight inherent in raw hour representations.
- 7) **Day-of-Year Sine and Cosine:** Cyclical encoding of the ordinal day using $\sin(2\pi d/365)$ and $\cos(2\pi d/365)$, capturing seasonal variation in solar declination and day length.

The prediction target is the normalised AC power output, computed as:

$$y_{\text{norm}} = \frac{P_{\text{AC}}}{P_{99}^{(i)}} \quad (1)$$

where P_{AC} denotes the measured AC power output and $P_{99}^{(i)}$ represents the 99th percentile AC power for inverter i , calculated from the training partition. This per-inverter normalisation strategy accommodates heterogeneous inverter capacities and efficiencies across the two plant installations.

C. Model Training and Hyperparameters

A Random Forest Regressor is trained with the hyperparameter configuration detailed in Table II.

TABLE II
RANDOM FOREST HYPERPARAMETER CONFIGURATION

Hyperparameter	Value
Number of estimators (trees)	200
Maximum tree depth	20
Minimum samples per leaf	2
Random state seed	42
Parallel jobs	-1 (all cores)

The dataset is partitioned temporally using an 80/20 ratio with a split timestamp of 2020-06-11T12:30:00 UTC, ensuring that the test set comprises exclusively future data unseen during model training. This temporal partitioning strategy is deliberately chosen over random splitting to faithfully simulate real-world deployment conditions where the model must predict future energy output based on patterns learned from historical observations.

D. Inference Pipeline

During inference, the system constructs feature vectors for each 15-minute interval spanning a full 24-hour day. Solar irradiance is estimated using a physics-informed model incorporating:

- Solar declination angle computed from the day of year
- Hour angle for sub-daily solar position variation
- Latitude-dependent zenith angle calculation

- Cloud cover attenuation factor: $f_c = \max(0.08, 1 - 0.75 \cdot c^{1.2})$
 - Humidity scattering factor: $f_h = \max(0.75, 1 - 0.20 \cdot h)$
- where c and h are the normalised cloud cover and humidity values, respectively. The model's normalised power predictions are then converted to absolute energy estimates:

$$E_{\text{daily}} = \sum_{t=1}^{96} \hat{y}_t \cdot C_p \cdot \eta_s \cdot \Delta t \quad (2)$$

where $\hat{y}_t$ is the predicted normalised power at interval t , C_p is the panel capacity (kW), $\eta_s = 0.82$ is the composite system loss factor accounting for inverter conversion losses, wiring resistance, and surface soiling, and $\Delta t = 0.25$ hours.

Panel efficiency is further modulated by temperature effects according to:

$$\eta_T = \eta_{\text{base}} \cdot (1 + k_T \cdot (T_{\text{cell}} - T_{\text{STC}})) \quad (3)$$

where $k_T = -0.004$ per $^{\circ}\text{C}$ is the temperature coefficient for crystalline silicon cells, $T_{\text{cell}} = T_{\text{ambient}} + 25^{\circ}\text{C}$ is the estimated cell temperature, and $T_{\text{STC}} = 25^{\circ}\text{C}$ is the standard test condition reference temperature.

V. GENERATIVE AI PLANNING SYSTEM AND DECISION SUPPORT FEATURES

A. Generative AI-Powered Planning and Advisory

A distinguishing feature of SolarPredict is its Generative AI-based planning system, which leverages the Google Gemini.

2.0 Flash a large language model to transform quantitative ML predictions into qualitative, actionable planning guidance. The AI planning module operates in two complementary modes:

- 1) **Automated Summary and Planning Mode:** Upon generation of prediction results, the system automatically constructs a structured prompt containing the user's location, weather data, energy predictions, financial analysis, and subsidy information. The Gemini LLM processes this context to generate a concise, non-technical narrative summarising the user's solar potential, estimated savings, environmental impact, and a practical implementation recommendation. This mode enables users without technical expertise to immediately understand the implications of the prediction output.
- 2) **Interactive Q&A Planning Mode:** Users can submit free-form natural language questions regarding any aspect of their solar investment—including savings projections, subsidy eligibility, ROI timelines, maintenance schedules, panel selection, and energy optimisation strategies. The system constructs a context-enriched prompt incorporating the user's prediction data and financial analysis, enabling the LLM to provide personalised, data-grounded responses rather than generic information.

A robust fallback mechanism utilising rule-based response templates ensures uninterrupted service continuity when the Generative AI API encounters rate limiting, network timeouts, or temporary unavailability.

B. Financial Analysis Engine

The system incorporates a comprehensive financial analysis module tailored specifically to the Indian residential solar market. Installation costs are modelled as a function of system capacity using tiered per-kW rates ranging from INR 42,000 for systems exceeding 10 kW to INR 65,000 for 1 kW systems, reflecting current 2024–2025 market pricing. Government subsidies under the PM Surya Ghar scheme are calculated according to:

$$S = \begin{cases} 30,000 \cdot C & \text{if } C \leq 2 \text{ kW} \\ 60,000 + 18,000 \cdot (C - 2) & \text{if } 2 < C \leq 3 \text{ kW} \\ 78,000 & \text{if } C > 3 \text{ kW} \end{cases} \quad (4)$$

where S denotes the subsidy amount in INR and C represents the system capacity in kW.

The ROI computation incorporates the following parameters:

- State-specific electricity tariff rates for 15 Indian states
- 0.5% annual panel degradation factor
- 5% annual electricity rate escalation assumption
- INR 2,000 annual maintenance cost provision
- 25-year cumulative net savings projection
- CO₂ emission offset at 0.82 kg per kWh (Indian grid emission factor)

Users can optionally input their current monthly electricity bill for a personalised bill comparison analysis showing the estimated post-solar adoption bill, monthly savings, and payback acceleration.

C. Smart Appliance Scheduling

The planning system generates intelligent appliance usage schedules based on predicted peak solar generation windows. High-power appliances (washing machine, water heater, air conditioner) are recommended for operation during peak solar hours (typically 10:00 AM–3:00 PM), thereby maximising self-consumption of solar energy. Forecast-aware alerts proactively notify users to shift energy-intensive loads when the succeeding day’s predicted output deviates by more than 20% from the current day’s generation.

D. User Authentication and Prediction History

JWT-based authentication enables registered users to persist prediction results to MongoDB and access their complete prediction history. This longitudinal tracking capability facilitates seasonal performance comparison, trend analysis, and validation of prediction accuracy against observed generation data over time.

VI. EXPERIMENTAL RESULTS AND ANALYSIS

A. Model Performance Evaluation

The trained Random Forest model is evaluated on the temporally partitioned test set comprising 27,280 records. Table III presents the comprehensive performance metrics.

TABLE III
RANDOM FOREST MODEL PERFORMANCE METRICS

Metric	Value
Normalized R ² Score	0.8534
Normalized MAE	0.0319
Absolute R ² Score	0.8578
Absolute MAE (Watts)	39.28
Training Samples	109,196
Test Samples	27,280
Number of Inverters	44

The normalised R² of 0.8534 exceeds the predefined baseline threshold of 0.82, indicating that the model explains approximately 85% of the variance in normalised power output. The normalised MAE of 0.0319 is substantially below the acceptable threshold of 0.05, confirming tight prediction accuracy. These metrics collectively validate the model’s capacity to generalise across temporal boundaries and heterogeneous inverter configurations.

The absolute metrics further corroborate model quality: an R² of 0.8578 on raw wattage predictions and an MAE of 39.28 W demonstrate that prediction errors remain small relative to typical inverter output ranges spanning 0–2,000 W.

B. Feature Importance Analysis

Fig. 2 and Table IV present the feature importance scores derived from the trained Random Forest model using the mean decrease in impurity criterion.

TABLE IV
FEATURE IMPORTANCE SCORES FROM RANDOM FOREST MODEL

Feature	Importance
Irradiation	0.8312
Wind Speed	0.0636
Cloud Cover	0.0397
Ambient Temperature	0.0196
Humidity	0.0122
Day-of-Year Sine	0.0100
Day-of-Year Cosine	0.0100
Hour Sine	0.0088
Hour Cosine	0.0049

Irradiation dominates with an importance score of 0.8312, unequivocally confirming its role as the primary determinant of solar power output. This finding aligns with established photovoltaic theory, as the photocurrent generated by PV cells is directly proportional to incident irradiance. The secondary features—wind speed (0.0636), cloud cover (0.0397), and ambient temperature (0.0196)—collectively capture the atmospheric conditions that modulate panel performance through cooling effects, light attenuation, and temperature-dependent efficiency variations. The cyclical time features contribute modestly (combined 0.0337), as their temporal information is substantially encoded within the irradiation measurements themselves.

VII. CONCLUSION AND FUTURE WORK

This paper presented SolarPredict, a comprehensive weather-based solar energy prediction and Generative AI-powered planning system designed specifically for Indian residential consumers. The system integrates a Random Forest regression model, achieving an R^2 of 0.853 with real-time weather data, detailed financial analysis including government subsidy calculations, and a Generative AI planning module powered by Google Gemini for personalised advisory services and smart energy management recommendations. The three-tier microservice architecture ensures modularity, independent scalability, and maintainability of each system component.

The principal contributions of this work include: (1) a feature engineering pipeline that synthesises real weather measurements with derived atmospheric parameters and cyclical time encodings; (2) a per-inverter P99 normalisation strategy enabling model generalisation across heterogeneous inverter configurations; (3) integration of comprehensive financial modelling specific to the Indian market encompassing PM Surya Ghar subsidy calculations and state-wise tariff structures; and (4) a Generative AI-augmented planning system that translates technical ML predictions into actionable natural language recommendations for non-technical residential consumers.

Future research directions include: (a) evaluating additional ML architectures including Gradient Boosting Machines and Long Short-Term Memory (LSTM) networks for ensemble prediction; (b) incorporating satellite-derived irradiance data for broader geographic coverage; (c) adding battery storage optimization and net metering simulation capabilities; (d) developing a mobile application for wider accessibility; (e) implementing multi-turn conversational planning with memory-augmented Generative AI; and (f) conducting field validation against actual metered generation data from rooftop installations across diverse Indian climatic zones.

ACKNOWLEDGMENT

The authors gratefully acknowledge the use of the Solar Power Generation Dataset made publicly available on Kaggle [12]. The OpenWeather API was used for real-time meteorological data, and the Google Gemini API was employed for Generative AI-based planning and advisory services. The authors also thank the Department of Information Technology, Dhanalakshmi Srinivasan College of Engineering and Technology, for institutional support.

REFERENCES

- [1] Ministry of New and Renewable Energy, Government of India, "India's updated first Nationally Determined Contribution under Paris Agreement," 2023. [Online]. Available: <https://mnre.gov.in/>
- [2] Press Information Bureau, "PM Surya Ghar: Muft Bijli Yojana - Rooftop Solar Scheme," Government of India, 2024. [Online]. Available: <https://pmsuryaghar.gov.in/>
- [3] R. Ahmed, V. Sreeram, Y. Mishra, and M. D. Arif, "A review and evaluation of the state-of-the-art in PV solar power forecasting: techniques and optimisation," *Renewable and Sustainable Energy Reviews*, vol. 124, p. 109792, 2020.

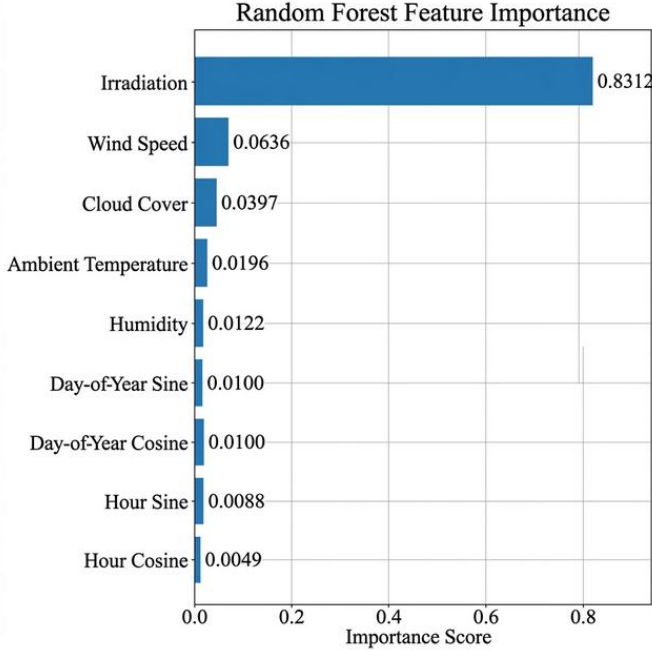


Fig. 2. Feature importance scores from the Random Forest model demonstrating the dominant contribution of irradiation (0.8312) in predicting solar power output, followed by secondary meteorological features.

C. System Performance Analysis

The complete prediction and planning pipeline—from weather API invocation through ML inference to Generative AI advisory generation—executes within 3–5 seconds under typical network conditions. The FastAPI ML inference service processes a single-day prediction request (96 feature vectors) in under 100 milliseconds on commodity hardware. The Express.js API gateway sustains 100 concurrent requests per 15-minute window with rate limiting, ensuring equitable resource allocation across concurrent users. The Generative AI planning component adds approximately 1–2 seconds of latency for LLM-based summary and recommendation generation, which is acceptable for the interactive use case.

D. Comparison with Existing Approaches

Table V provides a qualitative comparison of SolarPredict with existing solar estimation tools, highlighting the unique combination of ML-based prediction, Generative AI planning, and India-specific financial modelling.

TABLE V
COMPARISON WITH EXISTING SOLAR ESTIMATION TOOLS

Feature	SolarPredict	PVWatts	Sunroof
ML Prediction	✓	×	×
Gen AI Planning	✓	×	×
Real-time Weather	✓	×	×
India Subsidies	✓	×	×
Smart Scheduling	✓	×	×
Bill Comparison	✓	Partial	×
User History	✓	×	×

- [4] A. Mellit and A. M. Pavan, "A 24-h forecast of solar irradiance using artificial neural network: application for performance prediction of a grid-connected PV plant at Trieste, Italy," *Solar Energy*, vol. 84, no. 5, pp. 807–821, 2010.
- [5] M. Abuello and B. Chowdhury, "Forecasting of solar power ramp events: a post-processing approach," *Renewable Energy*, vol. 133, pp. 1380–1392, 2019.
- [6] H. C. Hottel, "A simple model for estimating the transmittance of direct solar radiation through clear atmospheres," *Solar Energy*, vol. 18, no. 2, pp. 129–134, 1976.
- [7] C. Voyant, G. Notton, S. Kalogirou, M.-L. Nivet, C. Paoli, F. Motte, and A. Fouilloy, "Machine learning methods for solar radiation forecasting: a review," *Renewable Energy*, vol. 105, pp. 569–582, 2017.
- [8] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [9] Google, "Project Sunroof – Solar savings estimator," 2023. [Online]. Available: <https://sunroof.withgoogle.com/>
- [10] National Renewable Energy Laboratory, "PVWatts Calculator," 2024. [Online]. Available: <https://pvwatts.nrel.gov/>
- [11] J. Chen, X. Liang, and Z. Li, "Large language models for energy systems: a survey," *arXiv preprint arXiv:2312.15719*, 2023.
- [12] Afroz, "Solar Power Generation Data," Kaggle, 2020. [Online]. Available: <https://www.kaggle.com/datasets/anikannal/solar-power-generation-data>
- [13] M. Abdel-Nasser and K. Mahmoud, "Accurate photovoltaic power forecasting models using deep LSTM-RNN," *Neural Computing and Applications*, vol. 31, no. 7, pp. 2727–2740, 2019.
- [14] J. Antonanzas, N. Osorio, R. Escobar, R. Urraca, F. J. Martinez-de-Pison, and F. Antonanzas-Torres, "Review of photovoltaic power forecasting," *Solar Energy*, vol. 136, pp. 78–111, 2016.
- [15] V. Kushwaha and N. M. Pindoriya, "A SARIMA-RVFL hybrid model assisted by wavelet decomposition for very short-term solar PV power generation forecast," *Renewable Energy*, vol. 140, pp. 124–139, 2019.
- [16] S. Sobri, S. Kooli-Kamali, and N. A. Rahim, "Solar photovoltaic generation forecasting methods: a review," *Energy Conversion and Management*, vol. 156, pp. 459–497, 2018.
- [17] P. Kumari and D. Toshniwal, "Deep learning models for solar irradiance forecasting: a comprehensive review," *Journal of Cleaner Production*, vol. 318, p. 128566, 2021.
- [18] H. T. C. Pedro and C. F. M. Coimbra, "Assessment of forecasting techniques for solar power production with no exogenous inputs," *Solar Energy*, vol. 86, no. 7, pp. 2017–2028, 2012.
- [19] M. Q. Raza, M. Nadarajah, and C. Ekanayake, "On recent advances in PV output power forecast," *Solar Energy*, vol. 136, pp. 125–144, 2016.
- [20] A. Srivastava and D. K. Seshadri, "An analysis of rooftop solar PV potential in Indian states using geographic information system," *Current Science*, vol. 121, no. 8, pp. 1052–1060, 2021.